Polylogarithmic Chabauty–Kim loci over number fields

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joint with X. Li

The plan

Chabauty–Kim over number fields

Polylogarithmic Chabauty–Kim loci for the thrice-punctured line

The theory behind

1. Chabauty–Kim over number fields

S-integral points on curves

- K number field, S finite set of primes of K
- $\mathcal{O}_{K,S} = \mathcal{O}_K[S^{-1}]$ ring of S-integers
- $X/\mathcal{O}_{K,S}$ regular model of smooth hyperbolic curve

Theorem (Siegel, Mahler, Faltings)

$\#X(\mathcal{O}_{K,S}) < \infty$

Example

$X = \mathbb{P}^1 \smallsetminus \{0, 1, \infty\}$, $K = \mathbb{Q}(i)$, $S = \{(1-i)\}$

$$\Rightarrow \; X\left(\mathbb{Z}[i, \tfrac{1}{1-i}]\right) = \left\{2, -1, \frac{1}{2}, i, -i, 1+i, 1-i, \frac{1+i}{2}, \frac{1-i}{2}\right\}$$

Chabauty–Kim over number fields

Fix

- a rational prime p which splits completely in K s.t. all $\mathfrak{p} \mid p$ are of good reduction and $\notin S$
- a G_K-equivariant quotient $\pi_1^{\text{ét},\mathbb{Q}_p}(X_{\overline{K}}, b) \twoheadrightarrow \Pi^{\text{ét}}$

Aim: locate $X(\mathcal{O}_{K,S})$ inside $X(\mathcal{O}_K \otimes_{\mathbb{Z}} \mathbb{Z}_p) \cong \prod_{\mathfrak{p}|p} X(\mathcal{O}_{\mathfrak{p}})$ or inside $X(\mathcal{O}_{\mathfrak{p}})$ for fixed $\mathfrak{p} \mid p$.

Chabauty–Kim method produces subsets (Chabauty–Kim loci)

$$X(\mathcal{O}_K \otimes_{\mathbb{Z}} \mathbb{Z}_p)_{S,\Pi^{\text{ét}}} \subseteq X(\mathcal{O}_K \otimes_{\mathbb{Z}} \mathbb{Z}_p),$$
$$X(\mathcal{O}_{\mathfrak{p}})_{S,\Pi^{\text{ét}}} \subseteq X(\mathcal{O}_{\mathfrak{p}})$$

containing $X(\mathcal{O}_{K,S})$.

Remark

When X is projective and $\Pi^{\text{ét}} = \pi_1^{\text{ét},\mathbb{Q}_p}(X_{\overline{K}}, b)^{\text{ab}}$, the CK method specialises to Siksek's Restriction of Scalars Chabauty.

Chabauty–Kim diagram

$$\begin{tikzcd} X(\mathcal{O}_{K,S}) \arrow[r, hook] \arrow[d, "j_S"] & X(\mathcal{O}_K \otimes_{\mathbb{Z}} \mathbb{Z}_p) = \prod_{\mathfrak{p}|p} X(\mathcal{O}_{\mathfrak{p}}) \arrow[d, "j_p = \prod_{\mathfrak{p}} j_{\mathfrak{p}}"] \\ \mathrm{Sel}_{S,\Pi^{\text{ét}}}(X) \arrow[r, "\mathrm{loc}_p = \prod_{\mathfrak{p}} \mathrm{loc}_{\mathfrak{p}}"] & \prod_{\mathfrak{p}|p} \mathrm{H}^1_f(G_{\mathfrak{p}}, \Pi^{\text{ét}}) \end{tikzcd}$$

- $\mathrm{Sel}_{S,\Pi^{\text{ét}}}(X) \subseteq \mathrm{H}^1(G_K, \Pi^{\text{ét}})$ (global Selmer scheme) parametrises $\Pi^{\text{ét}}$-torsors with G_K-action satisfying local conditions
- $\mathrm{H}^1_f(G_{\mathfrak{p}}, \Pi^{\text{ét}})$ (local Selmer scheme) parametrises crystalline $\Pi^{\text{ét}}$-torsors with $G_{\mathfrak{p}}$-action
- j_S and $j_{\mathfrak{p}}$ (global and local Kummer map) form path torsors:

$$x \mapsto \pi_1^{\text{ét},\mathbb{Q}_p}(X_{\overline{K}}; b, x) \overset{\pi_1^{\text{ét},\mathbb{Q}_p}(X_{\overline{K}}, b)}{\times} \Pi^{\text{ét}}$$

- $\mathrm{loc}_{\mathfrak{p}}$ restricts G_K-action to $G_{\mathfrak{p}} \subseteq G_K$

Chabauty–Kim loci

$$\begin{tikzcd} X(\mathcal{O}_{K,S}) \arrow[r, hook] \arrow[d, "j_S"] & X(\mathcal{O}_K \otimes_{\mathbb{Z}} \mathbb{Z}_p) = \prod_{\mathfrak{p}|p} X(\mathcal{O}_{\mathfrak{p}}) \arrow[d, "j_p = \prod_{\mathfrak{p}} j_{\mathfrak{p}}"] \\ \mathrm{Sel}_{S,\Pi^{\text{ét}}}(X) \arrow[r, "\mathrm{loc}_p = \prod_{\mathfrak{p}} \mathrm{loc}_{\mathfrak{p}}"] & \prod_{\mathfrak{p}|p} \mathrm{H}^1_f(G_{\mathfrak{p}}, \Pi^{\text{ét}}) \end{tikzcd}$$

Fact: $\mathrm{loc}_{\mathfrak{p}}$ is a morphism of affine $\mathbb{Q}_p$-schemes

Definition (Chabauty–Kim loci)

$$X(\mathcal{O}_K \otimes_{\mathbb{Z}} \mathbb{Z}_p)_{S,\Pi^{\text{ét}}} := j_p^{-1}(\mathrm{loc}_p(\mathrm{Sel}_{S,\Pi^{\text{ét}}}(X)))$$

$$X(\mathcal{O}_{\mathfrak{p}})_{S,\Pi^{\text{ét}}} := \mathrm{pr}_{\mathfrak{p}}(X(\mathcal{O}_K \otimes_{\mathbb{Z}} \mathbb{Z}_p)_{S,\Pi^{\text{ét}}})$$

Here, $\mathrm{pr}_{\mathfrak{p}} \colon X(\mathcal{O}_K \otimes_{\mathbb{Z}} \mathbb{Z}_p) = \prod_{\mathfrak{p}'|p} X(\mathcal{O}_{\mathfrak{p}'}) \to X(\mathcal{O}_{\mathfrak{p}})$ is the projection map.

Weaker variant for single prime $\mathfrak{p} \mid p$:

$$X(\mathcal{O}_{\mathfrak{p}})'_{S,\Pi^{\text{ét}}} := j_{\mathfrak{p}}^{-1}(\mathrm{loc}_{\mathfrak{p}}(\mathrm{Sel}_{S,\Pi^{\text{ét}}}(X))) \supseteq X(\mathcal{O}_{\mathfrak{p}})_{S,\Pi^{\text{ét}}}$$

Kim's Conjecture over number fields

Conjecture 1 (Kim's Conjecture)

For sufficiently large quotients $\pi_1^{\text{ét},\mathbb{Q}_p}(X_{\overline{K}}, b) \twoheadrightarrow \Pi^{\text{ét}}$:

$$X(\mathcal{O}_K \otimes_{\mathbb{Z}} \mathbb{Z}_p)_{S,\Pi^{\text{ét}}} = X(\mathcal{O}_{K,S}).$$

Conjecture 2 (Kim's Conjecture for $\mathfrak{p} \mid p$)

For sufficiently large quotients $\pi_1^{\text{ét},\mathbb{Q}_p}(X_{\overline{K}}, b) \twoheadrightarrow \Pi^{\text{ét}}$:

$$X(\mathcal{O}_{\mathfrak{p}})_{S,\Pi^{\text{ét}}} = X(\mathcal{O}_{K,S}).$$

- Conjecture 1 $\Rightarrow$ Conjecture 2 for all $\mathfrak{p} \mid p$
- converse can fail

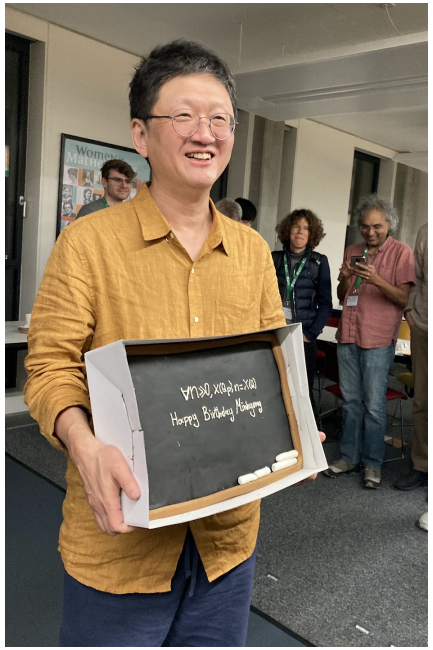


Photo credit: Jan Vonk

Relevance of Kim's Conjecture

Why do we care about Kim's Conjecture?

- reduces determination of $X(\mathcal{O}_{K,S})$ to computing CK loci
- implies a weak form of Grothendieck's Section Conjecture:

S-Selmer Section Conjecture

Every locally geometric S-integral Galois section $s\colon G_K \to \pi_1^{\text{ét}}(X, b)$ comes from an S-integral point.

Theorem (Betts–Kumpitsch–L., 2023)

If $X/\mathcal{O}_{K,S}$ satisfies Conjecture 2 for a Dirichlet-dense set of primes $\mathfrak{p}$ then it satisfies the S-Selmer Section Conjecture.

Dogra's finiteness results

- take $\Pi^{\text{ét}} = \Pi_n^{\text{ét}}$ the (nilpotency) depth n quotient
- locus $X(\mathcal{O}_K \otimes \mathbb{Z}_p)_{S,n}$ is cut out in $X(\mathcal{O}_K \otimes \mathbb{Z}_p) = \prod_{\mathfrak{p}|p} X(\mathcal{O}_\mathfrak{p})$ by iterated Coleman integrals of length $\leq n$
- at least $[K : \mathbb{Q}]$ functions needed to cut out a finite set
- need to rule out unlikely intersections

Theorem (Dogra, 2023)

If $X = \mathbb{P}^1 \smallsetminus \{0, 1, \infty\}$ then $\#X(\mathcal{O}_K \otimes_\mathbb{Z} \mathbb{Z}_p)_{S,n} < \infty$ for $n \gg 0$.

Theorem (Dogra, 2023)

Suppose X/K is smooth projective of genus > 1. Assume that the conjecture of Jannsen, or the conjecture of Bloch–Kato, hold for X^n for all n. Then $\#X(K \otimes_\mathbb{Q} \mathbb{Q}_p)_n < \infty$ for $n \gg 0$.

2. Polylogarithmic Chabauty–Kim loci for the thrice-punctured line

Polylogarithmic Chabauty–Kim loci

From now on:

- $X = \mathbb{P}^1 \smallsetminus \{0, 1, \infty\}$
- $\Pi^{\text{ét}} = \Pi^{\text{ét}}_{\mathrm{PL},n}$ depth-n polylogarithmic quotient $(1 \leq n \leq \infty)$

$\Rightarrow$ CK loci are cut out by polynomials in p-adic (poly)logarithms $\log, \mathrm{Li}_1, \ldots, \mathrm{Li}_n \colon X(\mathcal{O}_\mathfrak{p}) \to K_\mathfrak{p}$

$$\log(z) = \int_{\vec{1}_0}^{z} \frac{\mathrm{d}t}{t}, \quad \mathrm{Li}_k(z) = \int_{\vec{1}_0}^{z} \underbrace{\frac{\mathrm{d}t}{t} \cdots \frac{\mathrm{d}t}{t} \frac{\mathrm{d}t}{1-t}}_{k}$$

Notation for polylogarithmic Chabauty–Kim loci:

$$X(\mathcal{O}_{K,S} \otimes \mathbb{Z}_p)_{S,\mathrm{PL},n} \quad \text{resp.} \quad X(\mathcal{O}_\mathfrak{p})_{S,\mathrm{PL},n} \quad \text{resp.} \quad X(\mathcal{O}_\mathfrak{p})'_{S,\mathrm{PL},n}$$

What is known

Theorem (Kim, 2012)

If K is totally real then $\#X(\mathcal{O}_\mathfrak{p})'_{S,\mathrm{PL},n}$ is finite for $n \gg 0$.

Theorem (Dan-Cohen, Wewers, 2015)

For K a real quadratic field, $S = \emptyset$, $n = 2$, we have

$$X(\mathcal{O}_\mathfrak{p})'_{\emptyset,\mathrm{PL},2} = \left\{ z \in X(\mathbb{Z}_p) \;\middle|\; \mathrm{Li}_2(z) - \tfrac{1}{2}\log(z)\log(1-z) = 0 \right\}$$

Theorem (Dan-Cohen, 2020)

When K is not totally real then $X(\mathcal{O}_\mathfrak{p})'_{S,\mathrm{PL},\infty} = X(\mathcal{O}_\mathfrak{p})$.[1]

[1] except when K is imaginary quadratic and $S = \emptyset$

Integral points over imaginary quadratic fields

Theorem (Li–L.)

For K imaginary quadratic, $S = \emptyset$, any $1 \leq n \leq \infty$, the locus

$$X(\mathcal{O}_K \otimes \mathbb{Z}_p)_{\emptyset,\mathrm{PL},n}$$

is contained in the set of pairs $(z_1, z_2) \in X(\mathbb{Z}_p) \times X(\mathbb{Z}_p)$ satisfying

$$\begin{aligned} \log(z_1) = \log(z_2) &= 0, \\ \log(1 - z_1) = \log(1 - z_2) &= 0, \\ \mathrm{Li}_k(z_1) + (-1)^k \mathrm{Li}_k(z_2) &= 0 \text{ for } 2 \leq k \leq n. \end{aligned}$$

The containment is an equality assuming a p-adic version of Grothendieck's period conjecture.

Primitive sixth root of unity

We can solve the equations using the following lemma:

Lemma

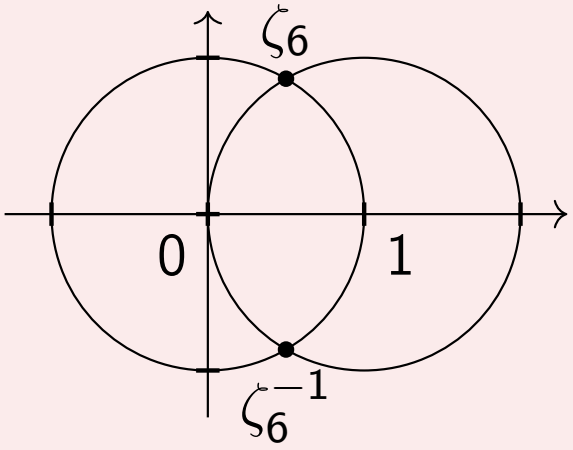
Let $z \in \mathbb{Z}_p$ *with* $z \not\equiv 0, 1 \bmod p$. *If* $\log(z) = 0$ *and* $\log(1-z) = 0$, *then* z *is a primitive* 6*-th root of unity.*

Proof.

If $\log(z) = 0$ and $\log(1-z) = 0$ then both z and $1-z$ are roots of unity. Fix an embedding of $\mathbb{Q}(z)$ into $\mathbb{C}$. Then

$$|z|_{\mathbb{C}} = 1 \text{ and } |1-z|_{\mathbb{C}} = 1.$$

The points of intersection of the unit circles centered at 0 and at 1 are $\zeta_6 = e^{2\pi i/6}$ and $\zeta_6^{-1} = e^{-2\pi i/6}$. □



Kim's Conjecture for $\mathbb{Z}[\zeta_3]$-points

The lemma implies:

$$X(\mathcal{O}_K \otimes \mathbb{Z}_p)_{\emptyset,1} = \mu_6^{\mathrm{prim}}(\mathbb{Z}_p) \times \mu_6^{\mathrm{prim}}(\mathbb{Z}_p)$$
$$X(\mathcal{O}_{\mathfrak{p}})_{\emptyset,1} = \mu_6^{\mathrm{prim}}(\mathbb{Z}_p)$$

Note that $X(\mathbb{Z}[\zeta_3]) = \{\zeta_6, \zeta_6^{-1}\}$ since $\zeta_6 + \zeta_6^{-1} = 1$.

Corollary

For $K = \mathbb{Q}(\zeta_3)$, $S = \emptyset$, any $p \equiv 1 \bmod 3$, $\mathfrak{p} \mid p$, Conjecture 2 holds in depth 1:

$$X(\mathcal{O}_{\mathfrak{p}})_{\emptyset,1} = X(\mathbb{Z}[\zeta_3]).$$

Kim's Conjecture for $\mathbb{Z}[\zeta_3]$-points

What about Conjecture 1?

A computation using finite polylogarithms shows $\mathrm{Li}_{p-3}(\zeta_6) \neq 0$ (Betts–Kumpitsch–L.)

$\Rightarrow (\zeta_6, \zeta_6), (\zeta_6^{-1}, \zeta_6^{-1}) \notin X(\mathcal{O}_K \otimes \mathbb{Z}_p)_{\emptyset,\mathrm{PL},p-3}$

$\Rightarrow X(\mathcal{O}_K \otimes \mathbb{Z}_p)_{\emptyset,\mathrm{PL},p-3} = \{(\zeta_6, \zeta_6^{-1}), (\zeta_6^{-1}, \zeta_6)\}$

Corollary

For $K = \mathbb{Q}(\zeta_3)$, $S = \emptyset$, any $p \equiv 1 \bmod 3$, Conjecture 1 holds in depth $p - 3$:

$$X(\mathbb{Z}[\zeta_3] \otimes \mathbb{Z}_p)_{\emptyset,\mathrm{PL},p-3} = X(\mathbb{Z}[\zeta_3]).$$

Remark: Conjecturally, $\mathrm{Li}_2(\zeta_6) \neq 0$, which would imply that depth 2 suffices.

Failure of polylogarithmic Kim's Conjecture

Theorem (Li–L.)

Let K be an imaginary quadratic field which is not $\mathbb{Q}(\zeta_3)$. Let p be a prime that splits completely in K and satisfies $p \equiv 1 \bmod 3$. Let $\mathfrak{p} \mid p$ and assume the p-adic period conjecture. Then

$$X(\mathcal{O}_K) = \emptyset \subsetneq \{\zeta_6, \zeta_6^{-1}\} = X(\mathcal{O}_{\mathfrak{p}})_{\emptyset,\mathrm{PL},\infty},$$

so Conjecture 2 fails for the polylogarithmic quotient.

S_3-symmetrisation

Corwin–Dan-Cohen observed a similar phenomenon over $\mathbb{Q}$ and proposed "S_3-symmetrisation" as a remedy: define

$$X(\mathcal{O}_{\mathfrak{p}})^{S_3}_{S,\mathrm{PL},n}$$

as the maximal S_3-stable subset of $X(\mathcal{O}_{\mathfrak{p}})_{S,\mathrm{PL},n}$. Note that

$$X(\mathcal{O}_{K,S}) \subseteq X(\mathcal{O}_{\mathfrak{p}})_{S,n} \subseteq X(\mathcal{O}_{\mathfrak{p}})^{S_3}_{S,\mathrm{PL},n} \subseteq X(\mathcal{O}_{\mathfrak{p}})_{S,\mathrm{PL},n}.$$

C–DC conjectured that $X(\mathcal{O}_{\mathfrak{p}})^{S_3}_{S,\mathrm{PL},n} = X(\mathcal{O}_{K,S})$ for $n \gg 0$.
But in the previous theorem, $X(\mathcal{O}_{\mathfrak{p}})_{\emptyset,\mathrm{PL},\infty} = \{\zeta_6, \zeta_6^{-1}\}$ is already S_3-stable, so even the S_3-symmetrised polylogarithmic locus is insufficient!

$\Rightarrow$ need to go beyond the polylogarithmic quotient eventually

Another negative result

Theorem (Li–L.)

Let K be a number field containing an imaginary quadratic field and assume $S \supseteq \{\mathfrak{l} \mid l\}$ for some rational prime l. Let $p \neq l$ be a prime which splits completely in K and assume the p-adic period conjecture. Then, for all $\mathfrak{p} \mid p$, the Chabauty–Kim locus $X(\mathcal{O}_{\mathfrak{p}})_{S,\mathrm{PL},\infty}$ contains all roots of unity $\neq 1$ in $\mathbb{Z}_p$.

When p is large, these roots of unity do not belong to $X(\mathcal{O}_{K,S})$, so Conjecture 2 fails for the polylogarithmic quotient.

Sets S of size 1, Galois-stable case

Variant of polylogarithms:

$$L_n(z) := \sum_{k=0}^{n-1} \frac{B_k}{k!} \log(z)^k \operatorname{Li}_{n-k}(z)$$

Example:

$$\begin{aligned}
L_1(z) &= \operatorname{Li}_1(z) = -\log(1-z), \\
L_2(z) &= \operatorname{Li}_2(z) - \tfrac{1}{2}\log(z)\operatorname{Li}_1(z), \\
L_3(z) &= \operatorname{Li}_3(z) - \tfrac{1}{2}\log(z)\operatorname{Li}_2(z) + \tfrac{1}{12}\log(z)^2\operatorname{Li}_1(z)
\end{aligned}$$

Sets S of size 1, Galois-stable case

Theorem (Li–L.)

Let K be an imaginary quadratic field, suppose $S = \{\mathfrak{l}\}$ is Galois-stable. Then $X(\mathcal{O}_K \otimes \mathbb{Z}_p)_{S,\mathrm{PL},4}$ is contained in the set of pairs $(z_1, z_2) \in X(\mathbb{Z}_p) \times X(\mathbb{Z}_p)$ satisfying

$$\begin{aligned}
\log(z_1) &= \log(z_2), \\
\log(1 - z_1) &= \log(1 - z_2), \\
L_2(z_1) + L_2(z_2) &= 0, \\
b_{\tau_\mathfrak{l}} b_{\sigma_2}(L_3(z_1) - L_3(z_2)) &= 2 b_{\tau_\mathfrak{l}\sigma_2} \log(z_1) L_2(z_1), \\
b_{\tau_\mathfrak{l}} b_{\sigma_3}(L_4(z_1) + L_4(z_2)) &= b_{\tau_\mathfrak{l}\sigma_3} \log(z_1)(L_3(z_1) + L_3(z_2)),
\end{aligned}$$

for certain constants $b_u \in \mathbb{Q}_p$.

Sets S of size 1, Galois-stable case

Equations for the depth 1 locus:

$$\log(z_1) = \log(z_2), \quad \log(1 - z_1) = \log(1 - z_2) \qquad (*)$$

Lemma

Two types of solutions to $(*)$:

- *diagonal:* (z, z), $z \in X(\mathbb{Z}_p)$
- *off-diagonal: pairs*

$$\left(\frac{1-\eta}{\zeta-\eta}, \zeta\frac{1-\eta}{\zeta-\eta}\right)$$

where ζ, η are roots of unity in $\mathbb{Z}_p$ with $\zeta, \eta \neq 1$ and $\zeta \neq \eta$

Sets S of size 1, Galois-stable case

Depth 1 locus $X(\mathcal{O}_K \otimes \mathbb{Z}_p)_{S,\mathrm{PL},1}$ is infinite due to diagonal solutions. Additional equation in depth 2:

$$L_2(z_1) + L_2(z_2) = 0$$

Corollary

The depth 2 locus $X(\mathcal{O}_K \otimes \mathbb{Z}_p)_{S,\mathrm{PL},2}$ is finite.

Theorem

Kim's Conjecture holds in depth 2 for $K = \mathbb{Q}(i)$, $S = \{(1-i)\}$, $p = 5$:

$$X(\mathbb{Z}[i] \otimes \mathbb{Z}_5)_{S,\mathrm{PL},2} = \left\{2, -1, \frac{1}{2}, i, -i, 1+i, 1-i, \frac{1+i}{2}, \frac{1-i}{2}\right\}.$$

Sets S of size 1, Galois-unstable case

Theorem (Li–L.)

Let K be an imaginary quadratic field, $S = \{\mathfrak{l}\}$ not Galois-stable. Then $X(\mathcal{O}_K \otimes \mathbb{Z}_p)_{S,\mathrm{PL},2}$ is contained in the set of pairs $(z_1, z_2) \in X(\mathbb{Z}_p) \times X(\mathbb{Z}_p)$ satisfying

$$\begin{aligned} \log^{\mathfrak{p}_2}(\alpha)\log(z_1) &= \log^{\mathfrak{p}_1}(\alpha)\log(z_2), \\ \log^{\mathfrak{p}_2}(\alpha)\log(1-z_1) &= \log^{\mathfrak{p}_1}(\alpha)\log(1-z_2), \\ L_2(z_1) + L_2(z_2) &= 0, \end{aligned}$$

where α is a generator of $\mathcal{O}_{K,S}^\times \otimes_{\mathbb{Z}} \mathbb{Q}$ and $\log^{\mathfrak{p}_i}$ is the $\mathfrak{p}_i$-adic logarithm on $K_{\mathfrak{p}_i} \cong \mathbb{Q}_p$.

A cyclotomic example

Theorem (Li–L.)

For $K = \mathbb{Q}(\zeta_8)$, $S = \{(1-\zeta_8)\}$, the depth-3 locus

$$X(\mathcal{O}_K \otimes \mathbb{Z}_p)_{S,\mathrm{PL},3}$$

is contained in the set of quadruples (z_1, z_2, z_3, z_4) satisfying

$$\begin{gathered}
X_1 = X_2, \quad X_3 = X_4, \\
Y_1 = Y_2, \quad Y_3 = Y_4, \\
Z_1 + Z_2 + Z_3 + Z_4 = 0, \\
Z_1 + Z_2 = c_1(-X_3 Y_1 + X_1 Y_3), \\
W_1 - W_2 - W_3 + W_4 + c_2(X_2 + X_4)(Z_2 + Z_3) + c_3(X_2 - X_4)(Z_2 + Z_4) = 0, \\
W_1 - W_2 + W_3 - W_4 + c_4(X_2 - X_4)(Z_2 + Z_3) + c_5(X_2 + X_4)(Z_2 + Z_4) = 0,
\end{gathered}$$

for some explicit constants $c_1, \ldots, c_5 \in \mathbb{Q}_p$, where

$$X_i := \log(z_i), \quad Y_i := L_1(z_i), \quad Z_i := L_2(z_i), \quad W_i := L_3(z_i).$$

A cyclotomic example

Remarks:

- Computationally, $\#X(\mathcal{O}_{\mathfrak{p}})_{S,\mathrm{PL},3} = 233$ for $\mathfrak{p} \mid p = 17$ but only 75 of them are S-integral points
- We have depth 4 equations but they contain coefficients which we cannot compute

Also have some results for:

- real quadratic fields
- refined Chabauty–Kim loci

3. The theory behind

The motivic Selmer scheme

To obtain equations for $X(\mathcal{O}_K \otimes \mathbb{Z}_p)_{S,\Pi^{\text{ét}}}$ we need to compute the scheme-theoretic image of the localisation map

$$\mathrm{Sel}_{S,\Pi^{\text{ét}}}(X) \xrightarrow{\prod_{\mathfrak{p}} \mathrm{loc}_{\mathfrak{p}}} \prod_{\mathfrak{p}|p} \mathrm{H}^1_f(G_{\mathfrak{p}}, \Pi^{\text{ét}}).$$

We work with the motivic Selmer scheme (Dan-Cohen, Wewers, Corwin, Brown, ...) and identify $\mathrm{loc}_{\mathfrak{p}}$ with a cocycle evaluation map.

The motivic Selmer scheme

- $\mathrm{MT}(\mathcal{O}_{K,S}, \mathbb{Q})$: category of mixed Tate motives over $\mathcal{O}_{K,S}$
- Deligne–Goncharov: motivic fundamental group $\pi_1^{\mathrm{mot}}(X, b)$ of $X = \mathbb{P}^1 \smallsetminus \{0, 1, \infty\}$ "in $\mathrm{MT}(\mathcal{O}_{K,S}, \mathbb{Q})$"
- fix quotient $\pi_1^{\mathrm{mot}}(X, b) \twoheadrightarrow \Pi$ in $\mathrm{MT}(\mathcal{O}_{K,S}, \mathbb{Q})$
- motivic Selmer scheme $\mathrm{Sel}^{\mathrm{mot}}_{S,\Pi}(X)$: affine $\mathbb{Q}$-scheme parametrising Π-torsors in $\mathrm{MT}(\mathcal{O}_{K,S}, \mathbb{Q})$
- comparison theorem (Betts–Kumpitsch–L.): étale realisation induces

$$\mathrm{Sel}^{\mathrm{mot}}_{S,\Pi}(X)_{\mathbb{Q}_p} \cong \mathrm{Sel}_{S,\Pi^{\text{ét}}}(X)$$

The motivic Selmer scheme

- have a canonical fibre functor $\omega\colon \mathsf{MT}(\mathcal{O}_{K,S},\mathbb{Q}) \to \mathrm{Vect}_{\mathbb{Q}}$:

$$\omega(M) = \bigoplus_{n\in\mathbb{Z}} \mathsf{Hom}(\mathbb{Q}(-n), \mathsf{gr}^W_{2n}(M))$$

- Tannakian fundamental group:

$$G_S^{\mathsf{MT}} = U_S^{\mathsf{MT}} \rtimes \mathbb{G}_m$$

  with U_S^{MT} a free pro-unipotent group
- motivic Selmer scheme is isomorphic to a space of $\mathbb{G}_m$-equivariant cocycles:

$$\begin{aligned}\mathsf{Sel}^{\mathsf{mot}}_{S,\Pi}(X) &\cong \mathsf{H}^1(G_S^{\mathsf{MT}}, \Pi^\omega)\\ &\cong \mathsf{Z}^1(U_S^{\mathsf{MT}}, \Pi^\omega)^{\mathbb{G}_m}\\ &\cong \mathsf{Z}^1(\mathrm{Lie}(U_S^{\mathsf{MT}}), \mathrm{Lie}(\Pi^\omega))^{\mathbb{G}_m}\end{aligned}$$

The cocycle evaluation map

Now take $\Pi = \Pi_{\mathrm{PL}}$ the polylogarithmic quotient.

U_S^{MT} acts trivially on $\Pi_{\mathrm{PL}}^{\omega}$, so cocycles = homomorphisms.

Let $\mathfrak{p} \mid p$. Chatzistamatiou–Ünver construct a $\mathfrak{p}$-adic period element

$$\eta_{\mathfrak{p}} \in U_S^{\mathsf{MT}}(K_{\mathfrak{p}})$$

converting between two functorial splittings of the weight filtration.

Let $\varepsilon_{\mathfrak{p}} := \log(\eta_{\mathfrak{p}})$. We get an evaluation map

$$\mathsf{Sel}^{\mathsf{mot}}_{S,\mathrm{PL}}(X) \cong \mathsf{Hom}(\mathrm{Lie}(U_S^{\mathsf{MT}}), \mathrm{Lie}(\Pi_{\mathrm{PL}}^{\omega}))^{\mathbb{G}_m} \xrightarrow{\mathsf{ev}_{\varepsilon_{\mathfrak{p}}}} \mathrm{Lie}(\Pi_{\mathrm{PL}}^{\mathrm{dR}})_{K_{\mathfrak{p}}}$$

The cocycle evaluation map

Proposition ("localisation = evaluation")

There is a commutative diagram

$$\begin{tikzcd}
\mathsf{Hom}(\mathrm{Lie}(U_S^{\mathsf{MT}}), \mathrm{Lie}(\Pi_{\mathrm{PL}}^{\omega})_{\mathbb{Q}_p}^{\mathbb{G}_m} \arrow[r, "\mathsf{ev}_{\varepsilon_{\mathfrak{p}}}"] \arrow[d, "\simeq"] & \mathrm{Lie}(\Pi_{\mathrm{PL}}^{\mathrm{dR}})_{K_{\mathfrak{p}}} \arrow[d, "\simeq"] \\
\mathsf{Sel}_{S,\mathrm{PL}}(X) \arrow[r, "\mathsf{loc}_{\mathfrak{p}}"] & \mathsf{H}_f^1(G_{\mathfrak{p}}, \Pi_{\mathrm{PL}}^{\text{ét}})
\end{tikzcd}$$

Coordinates on the Selmer scheme

Structure of $\mathrm{Lie}(\Pi^{\omega}_{\mathrm{PL}})$:

- $\mathrm{Lie}(\pi_1^{\omega}(X, b))$ is freely generated by e_0, e_1
- polylogarithmic quotient: mod out by nested commutators with more than one e_1
- basis of $\mathrm{Lie}(\Pi^{\omega}_{\mathrm{PL}})$: e_0 and $\mathrm{ad}(e_0)^{n-1}e_1$ $(n \geq 1)$

Structure of $\mathrm{Lie}(U_S^{\mathrm{MT}})$:

- free Lie algebra; get non-canonical generators by lifting basis from the abelianisation
- in degree $-n$:

$$\mathrm{Lie}(U_S^{\mathrm{MT}})^{\mathrm{ab}}_{-n} \cong \mathrm{Ext}^1_{\mathrm{MT}(\mathcal{O}_{K,S},\mathbb{Q})}(\mathbb{Q}(0), \mathbb{Q}(n))^{\vee} \cong K_{2n-1}(\mathcal{O}_{K,S})^{\vee}_{\mathbb{Q}}$$

- dimensions $d_n := \dim_{\mathbb{Q}} K_{2n-1}(\mathcal{O}_{K,S})_{\mathbb{Q}}$ known by Borel

Coordinates on the Selmer scheme

For $n \geq 1$, let $\sigma_{n,1}, \ldots, \sigma_{n,d_n} \in \mathrm{Lie}(U_S^{\mathrm{MT}})_{-n}$ be lifts of a basis from the abelianisation. Then the $(\sigma_{n,i})_{n,i}$ generate $\mathrm{Lie}(U_S^{\mathrm{MT}})$ freely. Notation: $\tau_i := \sigma_{1,i}$.

A graded homomorphism $c\colon \mathrm{Lie}(U_S^{\mathrm{MT}}) \to \mathrm{Lie}(\Pi_{\mathrm{PL}}^{\omega})$ is uniquely determined by

$$c(\tau_i) = x_i(c)e_0 + y_i(c)e_1 \quad (1 \leq i \leq d_1),$$
$$c(\sigma_{n,i}) = z_{n,i}(c)\,\mathrm{ad}(e_0)^{n-1}e_1 \quad (n \geq 2, 1 \leq i \leq d_n).$$

$\Rightarrow$ Selmer scheme is affine space

$$\mathrm{Sel}_{S,\mathrm{PL}}^{\mathrm{mot}}(X) \cong \mathrm{Spec}\,\mathbb{Q}[(x_i)_i, (y_i)_i, (z_{n,i})_{n,i}]$$

The cocycle evaluation map in coordinates

Proposition

Each element ε of $\mathrm{Lie}(U_S^{\mathrm{MT}})(R)$ for a $\mathbb{Q}$-algebra R can uniquely be written in the form

$$\varepsilon = \sum_{n=1}^{\infty} \varepsilon_n + \varepsilon_{\mathrm{Gon}}$$

where $\varepsilon_{\mathrm{Gon}}$ belongs to an ideal which is mapped to zero under any graded homomorphism to $\mathrm{Lie}(\Pi_{\mathrm{PL}}^{\omega})_R$, $\varepsilon_1 = \sum_{i=1}^{d_1} b_{\tau_i}\tau_i$, and

$$\varepsilon_n = \sum_{i_1 \leq \ldots \leq i_{n-1} > i_n} b_{\tau_{i_1}\cdots\tau_{i_n}} \operatorname{ad}(\tau_{i_1})\cdots\operatorname{ad}(\tau_{i_{n-1}})\tau_{i_n}$$
$$+ \sum_{m=2}^{n}\sum_{i=1}^{d_n}\sum_{i_1 \leq \ldots \leq i_{n-m}} b_{\tau_{i_1}\cdots\tau_{i_{n-m}}\sigma_{m,i}} \operatorname{ad}(\tau_{i_1})\cdots\operatorname{ad}(\tau_{i_{n-m}})\sigma_{m,i}$$

for $n \geq 2$, with coefficients $b_w \in R$, and all indices $i_j \in \{1, \ldots, d_1\}$.

The cocycle evaluation map in coordinates

Write the period element $\varepsilon_{\mathfrak{p}} \in \mathrm{Lie}(U_S^{\mathrm{MT}})_{K_{\mathfrak{p}}}$ in this form, with coefficients $b_w \in K_{\mathfrak{p}}$. We get explicit polynomials

$$f_n \in K_{\mathfrak{p}}[(x_i)_i, (y_i)_i, (z_{n,i})_{n,i}]$$

whose coefficients are linear combinations of the b_w, describing the cocycle evaluation map

$$\mathrm{Hom}(\mathrm{Lie}(U_S^{\mathrm{MT}}), \mathrm{Lie}(\Pi_{\mathrm{PL}}^{\omega}))_{\mathbb{Q}_p}^{\mathbb{G}_m} \xrightarrow{\mathrm{ev}_{\varepsilon_{\mathfrak{p}}}} \mathrm{Lie}(\Pi_{\mathrm{PL}}^{\mathrm{dR}})_{K_{\mathfrak{p}}},$$

$$c \mapsto c(\varepsilon_{\mathfrak{p}}) = \sum_{i=1}^{\infty} b_{\tau_i} x_i e_0 + \sum_{i=1}^{\infty} b_{\tau_i} y_i e_1 + \sum_{n=2}^{\infty} f_n(c)\, \mathrm{ad}(e_0)^{n-1} e_1$$

p-adic periods

Determining the coefficients $b_w \in K_{\mathfrak{p}}$ in the expansion of $\varepsilon_{\mathfrak{p}}$ is difficult!

The $\mathfrak{p}$-adic period conjecture implies that all b_w are $\neq 0$.

This includes e.g. the p-adic zeta values $\zeta(2n+1) \in \mathbb{Q}_p$, $n \geq 1$.

In degree -1: assume the basis $\tau_1, \ldots, \tau_{d_1}$ is dual to $\alpha_1, \ldots, \alpha_{d_1} \in \mathcal{O}_{K,S}^{\times} \otimes \mathbb{Q}$ under the isomorphism

$$\mathrm{Lie}(U_S^{\mathsf{MT}})_{-1} = \mathrm{Lie}(U_S^{\mathsf{MT}})_{-1}^{\mathsf{ab}} \cong K_1(\mathcal{O}_{K,S})_{\mathbb{Q}}^{\vee} \cong (\mathcal{O}_{K,S}^{\times} \otimes \mathbb{Q})^{\vee}.$$

Then

$$\varepsilon_{\mathfrak{p}} = b_{\tau_1}\tau_1 + \ldots + b_{\tau_{d_1}}\tau_{d_1} + \ldots$$

with

$$b_{\tau_i} = \log^{\mathfrak{p}}(\alpha_i) \in K_{\mathfrak{p}}.$$

Galois action

We get equations for the image of

$$\mathrm{Hom}(\mathrm{Lie}(U_S^{\mathrm{MT}}), \mathrm{Lie}(\Pi_{\mathrm{PL}}^{\omega}))_{\mathbb{Q}_p}^{\mathbb{G}_m} \xrightarrow{\prod_{\mathfrak{p}|p} \mathrm{ev}_{\varepsilon_{\mathfrak{p}}}} \prod_{\mathfrak{p}|p} \mathrm{Lie}(\Pi_{\mathrm{PL}}^{\mathrm{dR}})_{K_{\mathfrak{p}}}$$

by relating the various $\varepsilon_{\mathfrak{p}}$ ($\mathfrak{p} \mid p$) to each other.

Assume that $K/\mathbb{Q}$ is Galois and S is Galois-stable
$\Rightarrow$ $\mathrm{Gal}(K/\mathbb{Q})$ acts on U_S^{MT}, hence on $\mathrm{Lie}(U_S^{\mathrm{MT}})$

We have

$$\varepsilon_{\sigma(\mathfrak{p})} = \sigma(\varepsilon_{\mathfrak{p}})$$

Choose the generators $\sigma_{n,i}$ in such a way that we know how $\mathrm{Gal}(K/\mathbb{Q})$ acts on them, using

$$\mathrm{Lie}(U_S^{\mathrm{MT}})_{-n}^{\mathrm{ab}} \cong K_{2n-1}(\mathcal{O}_{K,S})_{\mathbb{Q}}^{\vee}.$$

Then the expansion of $\varepsilon_{\sigma(\mathfrak{p})}$ can be obtained from that of $\varepsilon_{\mathfrak{p}}$.

Example: imaginary quadratic fields

Let K be an imaginary quadratic field, $S = \emptyset$.
Let $\sigma \in \mathrm{Gal}(K/\mathbb{Q})$ be complex conjugation.
$\dim_{\mathbb{Q}} K_{2n-1}(K)_{\mathbb{Q}} = 1$ for all $n \geq 2$, whereas $\dim_{\mathbb{Q}} K_{2n-1}(\mathbb{Q})_{\mathbb{Q}} = 0$ for n even.
$\Rightarrow \sigma$ acts as $(-1)^{n+1}$ on $K_{2n-1}(K)_{\mathbb{Q}}$
$\Rightarrow$ can choose generator $\sigma_n \in \mathrm{Lie}(U_S^{\mathrm{MT}})_{-n}$ such that

$$\sigma(\sigma_n) = (-1)^{n+1}\sigma_n$$

Let p be totally split, $p\mathcal{O}_K = \mathfrak{p}_1\mathfrak{p}_2$

$$\begin{aligned} \varepsilon_{\mathfrak{p}_1} &= b_2\sigma_2 + b_3\sigma_3 + b_4\sigma_4 + \dots \\ \Rightarrow\ \varepsilon_{\mathfrak{p}_2} &= -b_2\sigma_2 + b_3\sigma_3 - b_4\sigma_4 \pm \dots \end{aligned}$$

Example: imaginary quadratic fields

Let $c\colon \mathrm{Lie}(U_S^{\mathrm{MT}}) \to \mathrm{Lie}(\Pi_{\mathrm{PL}}^{\omega})$ be a graded homomorphism.
Coordinates $z_n = z_n(c)$ defined by $c(\sigma_n) = z_n \operatorname{ad}(e_0)^{n-1} e_1$.

$$c(\varepsilon_{\mathfrak{p}_1}) = \sum_{n=2}^{\infty} b_n z_n \operatorname{ad}(e_0)^{n-1} e_1,$$

$$c(\varepsilon_{\mathfrak{p}_2}) = \sum_{n=2}^{\infty} (-1)^{n+1} b_n z_n \operatorname{ad}(e_0)^{n-1} e_1$$

Let $L_0, L_1, L_2, \ldots \in \mathrm{Lie}(\Pi_{\mathrm{PL}}^{\omega})^{\vee}$ be the linear duals of $e_0, e_1, [e_0, e_1], \ldots$, and let $L_n^{(i)} := L_n \circ \mathrm{pr}_i$ $(i = 1, 2)$.

The image of $\mathrm{ev}_{\varepsilon_{\mathfrak{p}_1}} \times \mathrm{ev}_{\varepsilon_{\mathfrak{p}_2}}$ is contained in the subspace of $\mathrm{Lie}(\Pi_{\mathrm{PL}}^{\omega}) \times \mathrm{Lie}(\Pi_{\mathrm{PL}}^{\omega})$ defined by $L_0^{(1)} = L_0^{(2)} = L_1^{(1)} = L_1^{(2)} = 0$ and

$$L_n^{(1)} + (-1)^n L_n^{(2)} = 0$$

Example: imaginary quadratic fields

We obtain the theorem mentioned earlier:

Theorem (Li–L.)

For K imaginary quadratic, $S = \emptyset$, any $1 \leq n \leq \infty$, the locus

$$X(\mathcal{O}_K \otimes \mathbb{Z}_p)_{\emptyset,\mathrm{PL},n}$$

is contained in the set of pairs $(z_1, z_2) \in X(\mathbb{Z}_p) \times X(\mathbb{Z}_p)$ satisfying

$$\begin{aligned} \log(z_1) = \log(z_2) &= 0, \\ \log(1 - z_1) = \log(1 - z_2) &= 0, \\ \mathrm{Li}_k(z_1) + (-1)^k \mathrm{Li}_k(z_2) &= 0 \text{ for } 2 \leq k \leq n. \end{aligned}$$

Thanks for listening